Yizheng Xie

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Education

Brown University, Providence, RI, USA — Sept 2023 – Present
Ph.D. Candidate in Computer Science (Distributed System, Operating System, Cloud Computing)

Boston University, Boston, MA, USA — Sept 2021 – May 2023
M.S. in Computer Science

Chinese University of Hong Kong, Shenzhen, Shenzhen, Guangdong, China — Sept 2017 – May 2021
B.E. in Electronic Information Engineering

Research

Slowpoke: End-to-end Throughput Optimization Modeling for Microservices — Sept 2023 – Apr 2025
To appear at NSDI'26 (Microservice, Distributed Tracing, Kubernetes, Service Mesh, Docker, Golang, C++)

- Propose a novel linear programming-based performance model to quantify end-to-end throughput improvements of hypothetical microservice optimizations without requiring white-box knowledge.
- Develop the first system to accurately predict throughput in complex microservices, achieving RMSE of 2.1% across 4 real-world benchmarks and 1.9% across 108 synthetic benchmarks with diverse characteristics.
- Design a lightweight distributed slowdown mechanism enabling coordinated pausing across large-scale microservices.

λEASH: Scaling Out Shell Scripts with Resumable Cloud Computing — Sept 2023 – Jan 2025
Under submission of EuroSys'26 (Serverless, AWS Lambda, Shell, Python, C, Rust)

- Develop a system to scale out unmodified shell scripts using cloud (serverful and serverless) computing, achieving 12.7× average and 154.2× peak speedups over Bash with high cost efficiency; in some cases both faster and cheaper.
- Design a *suspend-and-resume* protocol that extends AWS Lambda beyond the 15-minute execution limit while preserving correctness and Unix-style streaming efficiency, with negligible slowdown (avg: 0.03×).
- Propose a hybrid scheduler to balance workloads across EC2 and Lambda, enabling direct TCP communication between Lambdas via hole-punching and automatically resolving software dependencies despite bandwidth constraints.

Fractal: Fault-Tolerant Shell-Script Distribution — Sept 2023 – Dec 2024
To appear at NSDI'26 (Fault-Tolerance, Hadoop Distributed File System, Shell, Golang, C)

- Develop a system for fault-tolerant distribution of unmodified shell scripts, delivering average speedups of 9.6× over Bash, 5.5× over Hadoop Streaming (AHS), and 17% over DISH, the state-of-the-art shell scaleout system.
- Design a lightweight fault-recovery mechanism—including remote-pipe instrumentation and dynamic result persistence—that achieves recovery within 1.26× of fault-free execution and delivers a 9.3× speedup over AHS.
- Introduce a large-scale fault-injection tool to precisely characterize recovery behavior under real-world conditions.

Working Experience

Meta Platforms, Menlo Park, CA — May 2022 – Aug 2022
Production Engineer Intern in Kernel Team (Python, Fedora, Rust Packaging, Upstream Userspace)

- Integrate Fedora and CentOS Stream forges and issue trackers (Pagure, GitLab, Bugzilla) into Meta Internal Task tool.
- Improve open-source tools, e.g., fixing Bugzilla authentication issues and adding Copr build integration in review tool.
- Implement a CLI tool to automate the process of updating Rust RPM packages by recursively updating dependencies and packaging missing dependencies, analyzing parallel chain-build order and checking compatibility for the current update set or any Bodhi rust update set. (Pagure Repository: pagure.io/fedora-rust/rust-update-set)

Publications

[1] Zhicheng Huang, Ramiz Dundar, Yizheng Xie, Konstantinos Kallas, and Nikos Vasilakis. Fractal: Fault-tolerant shell-script distribution. In *23rd USENIX Symposium on Networked Systems Design and Implementation*, Renton, WA, May 2026. USENIX Association.

[2] Yizheng Xie, Di Jin, and Nikos Vasilakis. Towards hybrid cooperative-preemptive scheduling. In *Proceedings of the 13th Workshop on Programming Languages and Operating Systems*, Seoul, Republic of Korea, October 2025. ACM.

[3] Yizheng Xie, Di Jin, Oğuzhan Çölkesen, Vasiliki Kalavri, John Liagouris, and Nikos Vasilakis. Slowpoke: End-to-end throughput optimization modeling for microservice applications. In *23rd USENIX Symposium on Networked Systems Design and Implementation*, Renton, WA, May 2026. USENIX Association.